

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								
	1	2	3	4	5	6	7	8

1. Connector & input protection

Field connector, 2-stage bidirectional TVS surge protection, chassis isolation

File: connector.kicad_sch

2. Power

eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V

File: power.kicad_sch

3. MCU

STM32G0B1CCT3, SWD, status LED, fault pull-ups

File: mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04) and Pt1000 4-wire (ADS1220)

File: measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611 miswiring protection

File: analog_out.kicad_sch

6. RS-485

THVD2450 +/-70V fault-protected transceiver

File: rs485.kicad_sch

DOTECH HMT500 – Oil Moisture Transmitter – Schematic v0.3

v0.3: MCU STM32G0B1CCT3 (LQFP48, -40..125 C, in-house part); TPS26600PWP official pinout; single axial PCB 57x23 (mech Rev C).

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2450 -> J1

Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-30 V DC.

Protection target: any pin pair +/-30 V continuous (miswiring), surge +/-1 kV, ESD +/-8 kV contact.

Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), VDDA, CHASSIS. Inter-sheet signals: global labels.

VERIFY: LMR36006, PCAP04, DAC8760, TPS26611 pin numbers are placeholders (datasheets not accessible when drawn). TPS2660: RTN link R22 to confirm.

TBD footprints: M Connect 8P, CMC, GDT, feedthrough, buck inductor, LMR36006, PCAP04, DAC8760, TPS26611.

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22

Design notes: docs/hw/circuit-design.md

Oil moisture transmitter HMT500 – RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

Sheet: /

File: HMT500.kicad_sch

Title: HMT500 Top

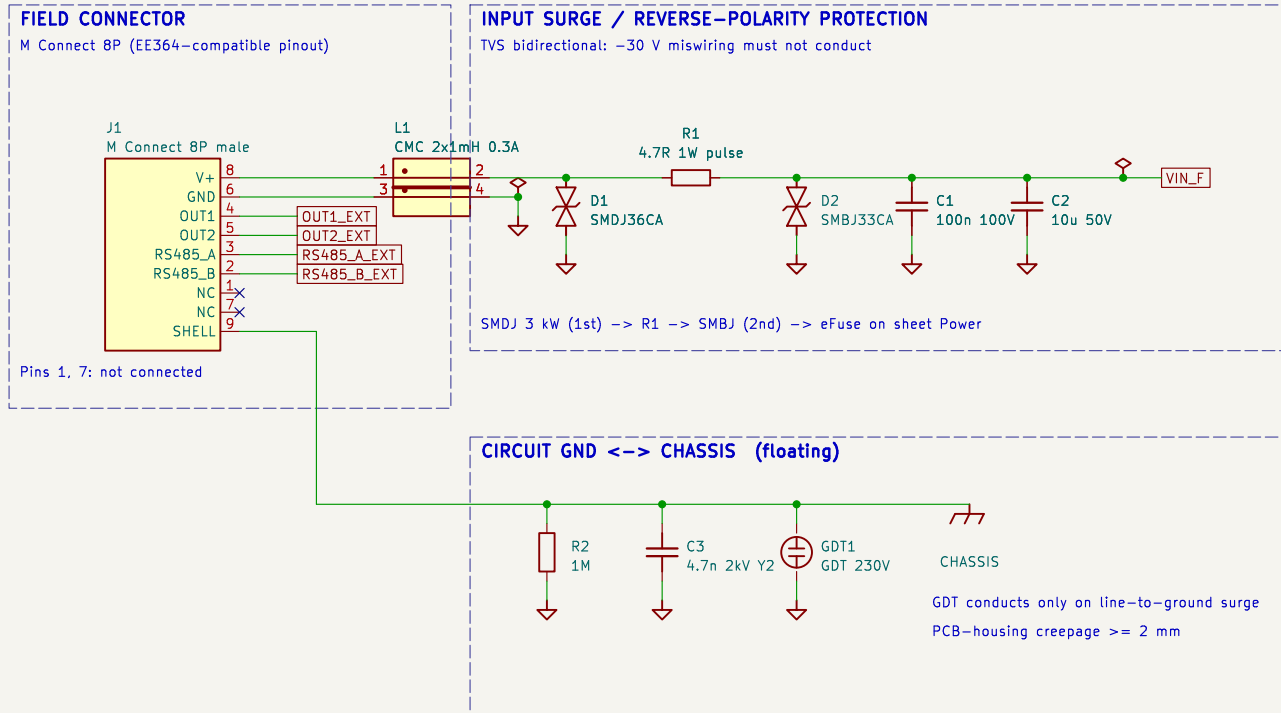
Size: A3

Date: 2026-09-26

Rev: 0.3

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 1/7



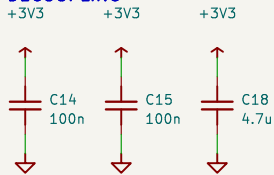
VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output
DOTECH Co., Ltd.

Sheet: /1. Connector & input protection/
File: connector.kicad_sch

Title: HMT500 Connector & input protection

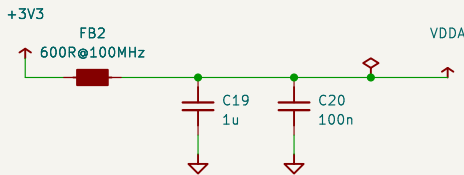
Size: A4	Date: 2026-09-26	Rev: 0.3
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 2/7

MCU DECOUPLING



100 nF at VDD and VBAT, 4.7 uF bulk

VDDA FILTER



MCU NOTES (STM32G0B1)

BOOT0 shares PA14/SWCLK – no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1

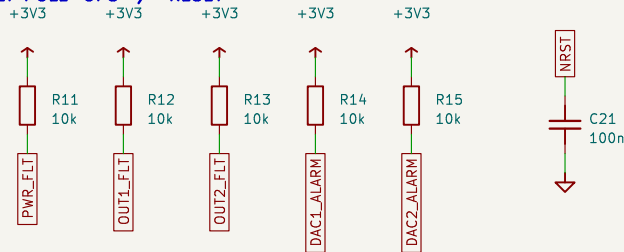
-> boot from flash (RS-485 bootloader 0x08000000).

VDDA bonded to VDD in LQFP48; VREF+ via FB2.

Cortex-M0+: no SWO (J2 pin 6 NC).

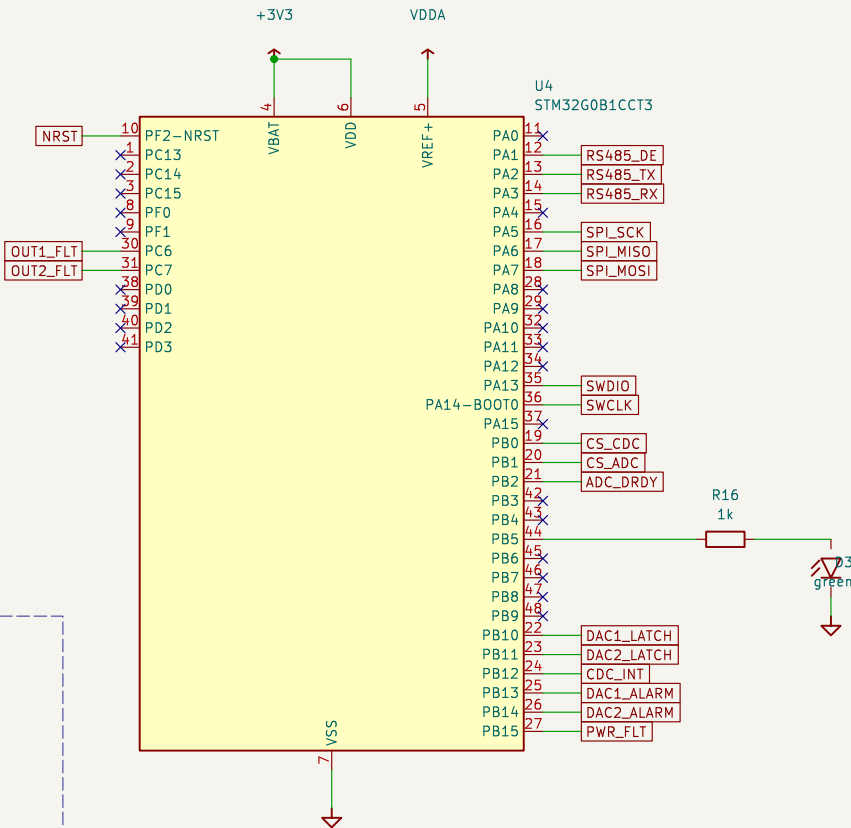
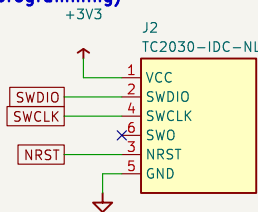
Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

FAULT PULL-UPS / RESET



Open-drain fault lines pulled up to +3V3

SWD (production programming)



VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 – RS-485 + 2x V/I analog output
DOTECH Co., Ltd.
Sheet: /3. MCU/
File: mcu.kicad_sch

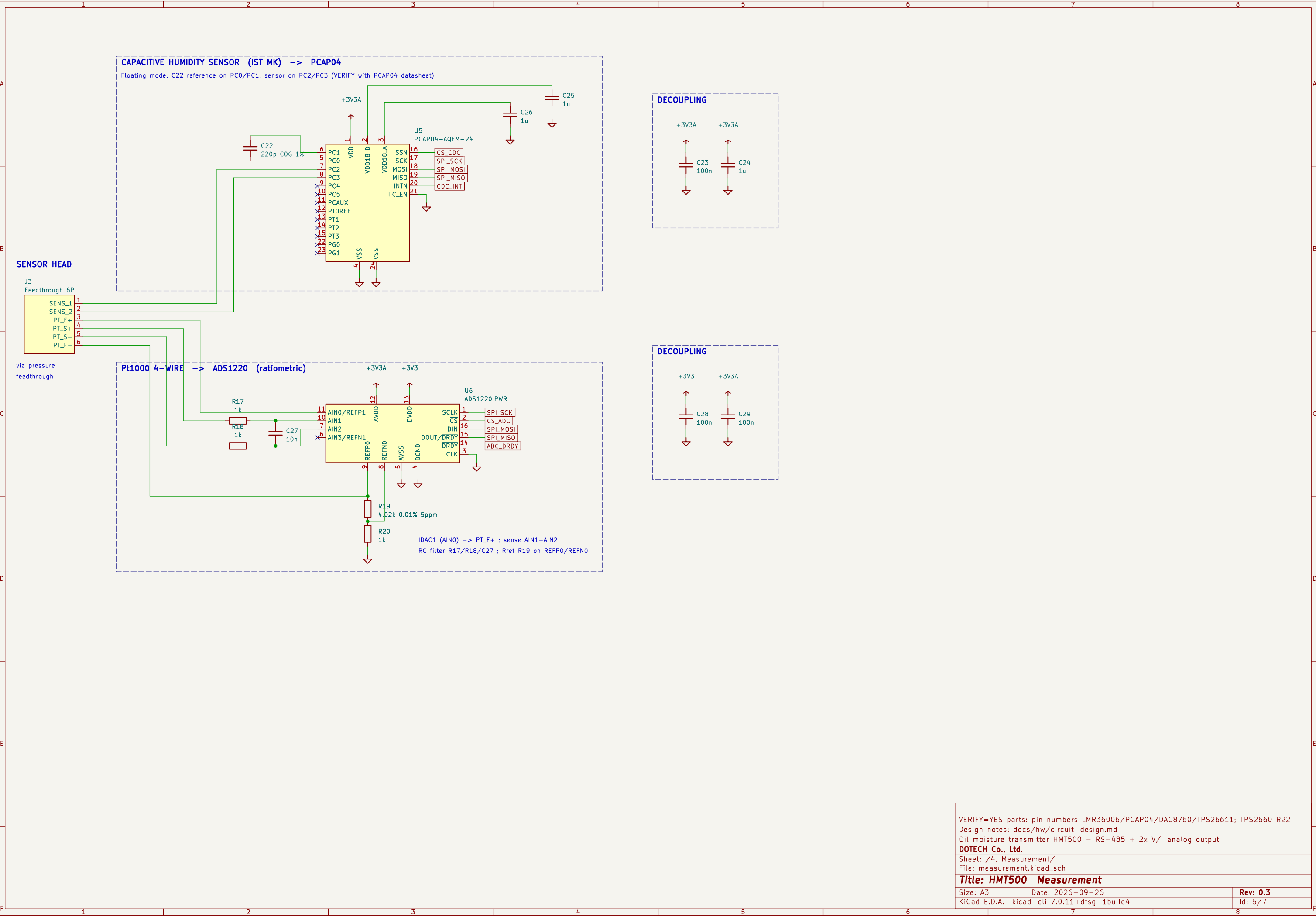
Title: HMT500 MCU

Size: A3 Date: 2026-09-26

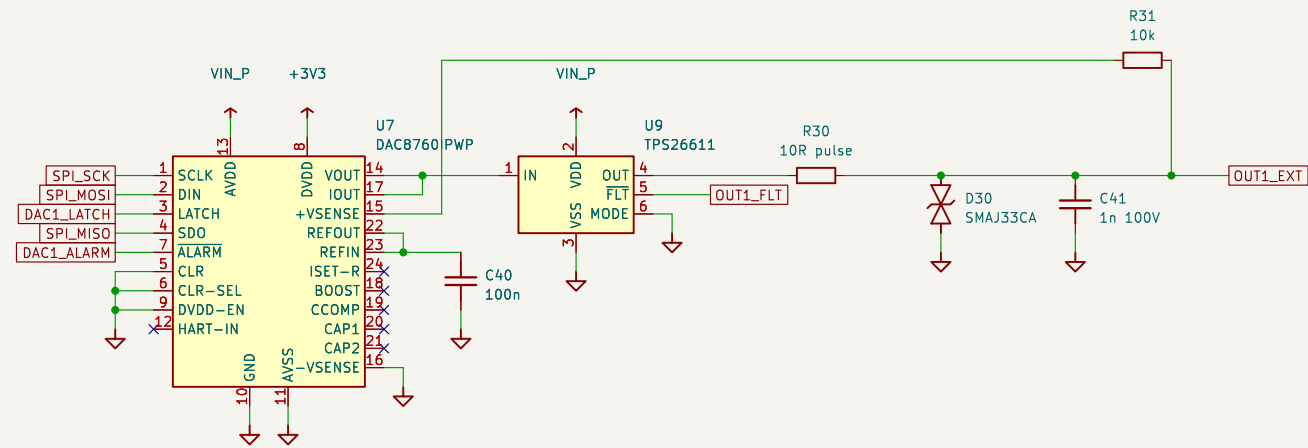
Rev: 0.3

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 4/7

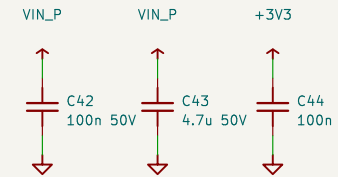


ANALOG OUTPUT CH1 - V/I selectable, miswiring protected -> connector pin 4

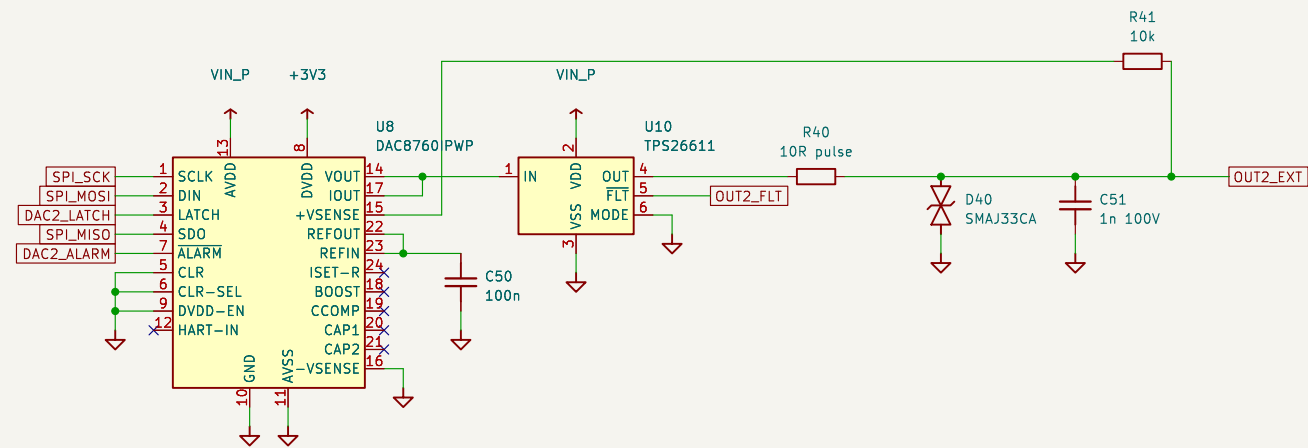


| VOUT and IOUT combined (per TI TIDUBK2 – VERIFY). +VSENSE taken after R30 to cancel series drop.

CH1 DECOUPLING

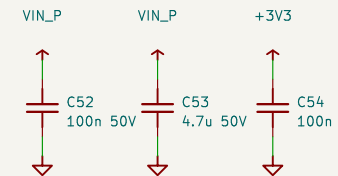


ANALOG OUTPUT CH2 - V/I selectable, miswiring protected -> connector pin 5



VOUT and IOOUT combined (per TI TIDUBK2 – VERIFY). +VSENSE taken after R40 to cancel series drop.

CH2 DECOUPLING



DAC SELECTION (decision 2026-09-26: fit DAC8760 now; final part after output spec is fixed)

U7/U8 fitted: DAC8760 (16-bit, current + voltage output). Same package/pinout family – drop-in alternates, no layout change:

DAC7760 12-bit, current + voltage (cost-down if V/I selectable output is kept)

DAC8750 16-bit, current only 4-20/0-20/0-24 mA (EE364-equivalent, 3-wire)

DAC7750 12-bit, current only (lowest cost; ~5 uA/step vs EE364 2 uA)

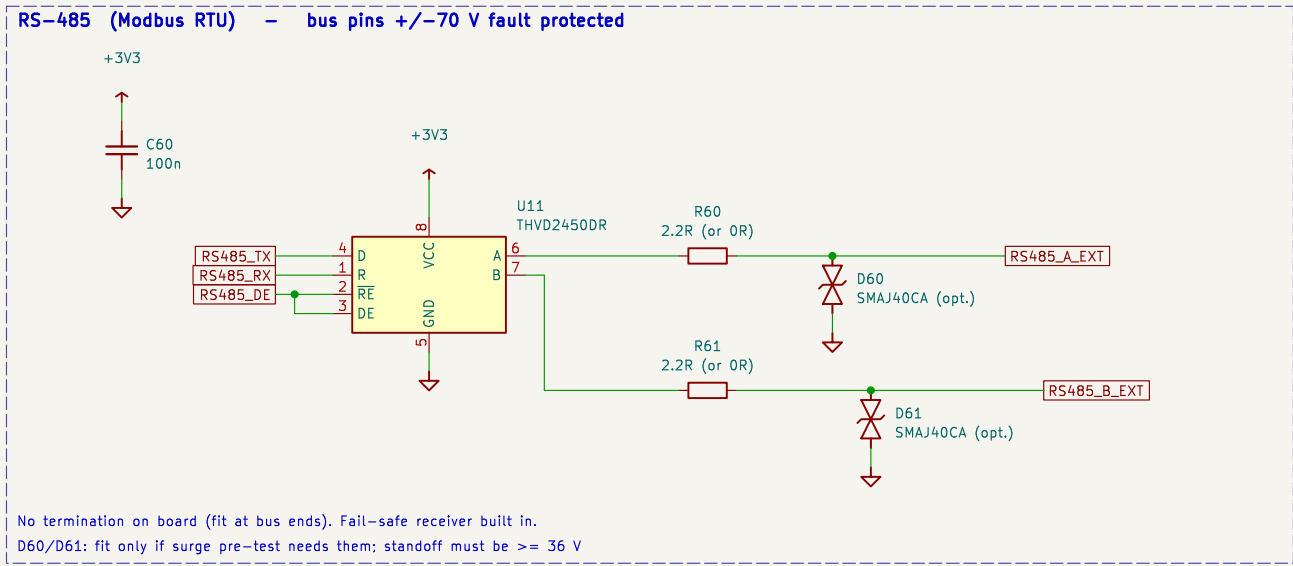
Current-only (x750): VOUT/+VSENSE/-VSENSE functions absent – check pin handling in datasheet before fitting (VERIFY).

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output
DOTECH Co., Ltd.

Sheet: /5. Analog outputs/
File: analog_out.kicad_sch

Title: HMT500 Analog outputs

Size: A3	Date: 2026-09-26	Rev: 0.3
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/7



VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

Sheet: /6. RS-485/
File: rs485.kicad_sch

Title: HMT500 RS-485

Size: A4	Date: 2026-09-26	Rev: 0.3
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/7